Fundamentals of Linear Algebra for AI

The Mathematical Backbone of Machine Learning

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Refrences

- "Introduction to Linear and Matrix Algebra", Nathaniel Johnston,
- "An Introduction to Linear Algebra for Science and Engineering", Daniel Norman, Dan Wolczuk,Third Edition

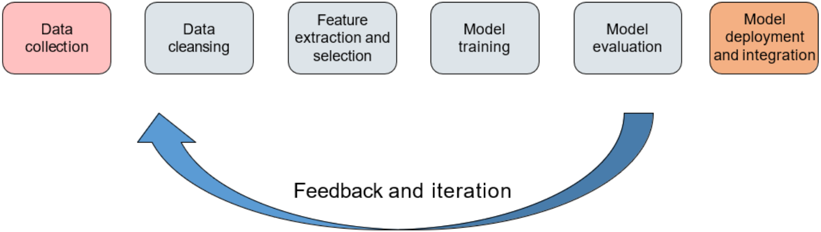
Table of Contents (I/V)

Introduction & Vectors

Why Linear Algebra for AI?

- **Data Representation:** All data (images, text, audio) is ultimately converted into numerical arrays (vectors/matrices).
- **Model Operations:** Machine Learning algorithms are sequences of matrix multiplications and vector additions.
- **Efficiency:** Linear Algebra operations are highly optimized for GPUs and modern hardware.

Data's Journey to the Model in the AI Pipeline



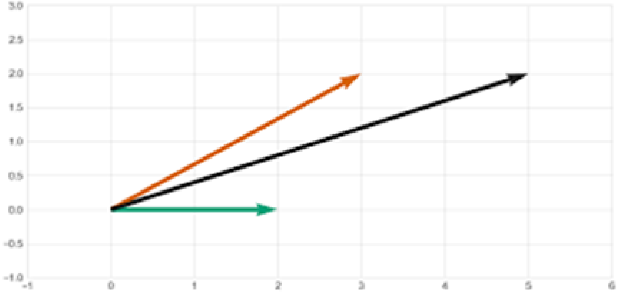
- Collection & Cleaning (Raw data stage)..
- Preparation & Structuring.
- Model Training (The algorithm stage).
- Inference.

4. Scalars, Vectors, and Tensors

- **Scalar (a):** A single number (0D array). E.g., learning rate.
- **Vector ($\mathbf{v}$):** An ordered list of numbers (1D array). E.g., feature vector.
- **Matrix ($\mathbf{A}$):** A 2D array of numbers. E.g., image data or weight layer.
- **Tensor ($\mathcal{T}$):** A generalization to N dimensions (N-D array). E.g., color video data ($W \times H \times C \times T$).

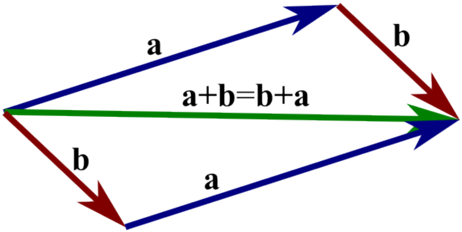
Vectors as Geometric Entities

- A vector $\mathbf{v} \in \mathbb{R}^n$ represents a point or a displacement in n-dimensional space.
- **Example:** A feature vector $\mathbf{x} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ is a point in $\mathbb{R}^2$.



Vector Operations: Addition

- Performed element-wise: $\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$.
- Geometrically, this is the **parallelogram rule**.
- **AI Context:** Combining feature effects.



Magnitude of A Vector

- A vector represents a direction and a magnitude.
- for $\vec{v} = v_1\hat{\imath} + v_2\hat{\jmath} + v_3\hat{k}$.
- magnitude $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$.

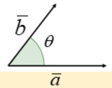
Vector Operations: Scaling

- Multiplies every element by the scalar c: $c\mathbf{v} = \begin{pmatrix} cv_1 \\ cv_2 \end{pmatrix}$.
- Geometrically, this scales the length of the vector.
- If $c < 0$, the vector's direction is reversed.

The Dot Product

The Dot Product (Inner Product)

- $\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$. The result is a scalar.
- $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}|| \cdot ||\mathbf{v}|| \cdot \cos(\theta)$.
- **Key property:** If $\mathbf{u} \cdot \mathbf{v} = 0$, the vectors are **orthogonal**.
- **AI Context:** Calculating similarity (Cosine Similarity) or weighted sums (Neural Networks).



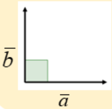
If θ is the angle between $\overline{a}$ and $\overline{b}$ then

$\overline{a} \bullet \overline{b} = |\overline{a}||\overline{b}|\cos\theta$

If $\overline{a} =< a_1, a_2, a_3 >$ and $\overline{b} =< b_1, b_2, b_3 >$

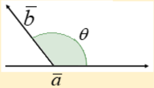
then the dot product is

$$\overline{a} \bullet \overline{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$



$\overline{a} \bullet \overline{b}$ are orthogonal (perpendicular)

if and only if $\overline{a} \bullet \overline{b} = 0$



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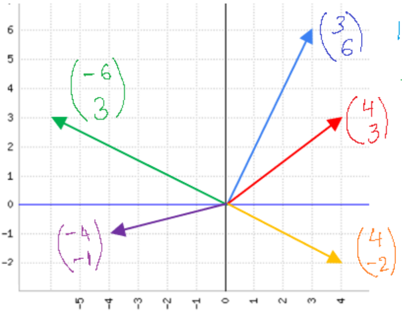
The Dot Product (Inner Product)

- $\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$. The result is a scalar.
- $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}|| \cdot ||\mathbf{v}|| \cdot \cos(\theta)$.
- It measures the projection of one vector onto another.

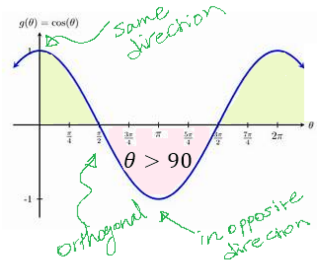


$12 + 18 = 30$

$-24 + 9 = -15$

$-16 - 3 = -18$

$16 - 6 = 10$



9. The Vector Norm (Length)

- The length or magnitude of a vector.
- **L2 Norm (Euclidean Norm):** $||\mathbf{v}||_2 = \sqrt{\sum_{i=1}^{n} v_i^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}}$.
- **L1 Norm (Manhattan Norm):** $||\mathbf{v}||_1 = \sum_{i=1}^{n} |v_i|$.
- **AI Context:** Used as a cost or regularization term (Lasso/Ridge Regression) to control model complexity.

10. Orthogonality and Basis Vectors

- Two vectors are orthogonal if the angle between them is 90° ($\cos(\theta) = 0$).
- **Unit Vector:** A vector with a norm of 1.
- **Orthonormal Basis:** A set of vectors that are all unit length and mutually orthogonal.
- **Importance:** Simplifies calculations and is key to transformations like PCA.

The Cross Product - Definition & Geometry

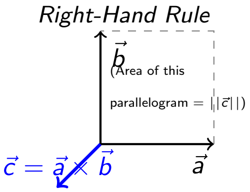
From two vectors to one new, orthogonal vector

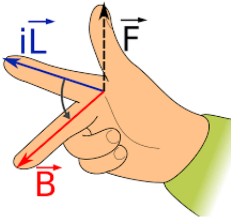
Definition

The **Cross Product** ($\vec{a} \times \vec{b}$) is a binary operation on two vectors in 3D space. The result, $\vec{c}$, is a new vector that is **orthogonal** (perpendicular) to both $\vec{a}$ and $\vec{b}$.

Geometric Interpretation

- **Direction:** Determined by the **Right-Hand Rule**.
- **Magnitude:** The magnitude of the resulting vector, $||\vec{a} \times \vec{b}||$, is equal to the **area of the parallelogram** formed by $\vec{a}$ and $\vec{b}$.





Cross product

The Cross Product as a Matrix Determinant

A practical method for calculation

The Formula

For $\vec{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, the cross product $\vec{a} \times \vec{b}$ is calculated as the determinant of this 3×3 matrix:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ -(a_1 b_3 - a_3 b_1) \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard basis vectors for the x, y, and z axes.

The Cross Product as a Matrix Determinant

A practical method for calculation

Numerical Example

Let $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} = \begin{pmatrix} (2 \cdot 6) - (3 \cdot 5) \\ (3 \cdot 4) - (1 \cdot 6) \\ (1 \cdot 5) - (2 \cdot 4) \end{pmatrix} = \begin{pmatrix} 12 - 15 \\ 12 - 6 \\ 5 - 8 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$$

This new vector is orthogonal to both $\vec{a}$ and $\vec{b}$.

Matrix Transformations - Rotation

Using matrices to rotate space

2D Rotation Matrix (R_θ)

To rotate a vector $\vec{v}$ counter-clockwise by an angle θ:

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

The new vector $\vec{v}'$ is: $\vec{v}' = R_\theta \cdot \vec{v}$

Matrix Transformations

Matrix Transformations - Rotation

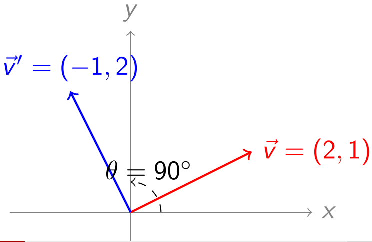
Using matrices to rotate space

Example: 90° Rotation

Let's rotate $\vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ by $\theta = 90^\circ$. $(\cos 90^\circ = 0, \sin 90^\circ = 1)$

$$R_{90^\circ} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\vec{v}' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} (0 \cdot 2) + (-1 \cdot 1) \\ (1 \cdot 2) + (0 \cdot 1) \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$



Matrix Transformations - Scaling

Using matrices to stretch and shrink space

Concept

A **Scaling Matrix** is a diagonal matrix that scales vectors along the standard axes.

2D Scaling Matrix (S)

To scale by s_x on the x-axis and s_y on the y-axis:

$$S = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}$$

- **Uniform Scaling:** $s_x = s_y$ (preserves shape)
- **Differential Scaling:** $s_x \neq s_y$ (stretches shape)

Matrix Transformations - Scaling

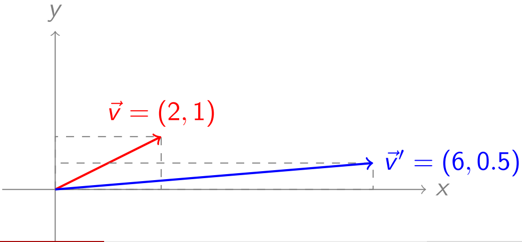
Using matrices to stretch and shrink space

Example: Stretch

Let's scale $\vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ by $s_x = 3$ and $s_y = 0.5$.

$$S = \begin{pmatrix} 3 & 0 \\ 0 & 0.5 \end{pmatrix}$$

$$\vec{v}' = \begin{pmatrix} 3 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} (3 \cdot 2) + (0 \cdot 1) \\ (0 \cdot 2) + (0.5 \cdot 1) \end{pmatrix} = \begin{pmatrix} 6 \\ 0.5 \end{pmatrix}$$



Singular Matrices & Degenerate Transformations

When transformations lose information

Singular Matrix

A square matrix is *singular* if its *determinant is 0*.

- This means the matrix has **no inverse**. You cannot "undo" its transformation.
- Geometrically, a singular matrix performs a **degenerate transformation**: it squashes N-dimensional space into a lower dimension.
- A 2D singular matrix collapses all vectors onto a single line or a single point.

Singular Matrices

Singular Matrices & Degenerate Transformations

When transformations lose information

Numerical Example: Projection

Let $M = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. $\det(M) = (1 \cdot 0) - (0 \cdot 0) = 0$. **This matrix is singular.**

Apply M to any vector $\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$:

$$\vec{v}' = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix}$$

Geometric Result

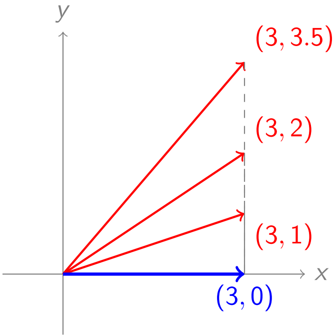
This transformation **projects all vectors onto the x-axis**. It's a

"many-to-one" mapping. For example, $(3, 2)$, $(3, 5)$, and $(3, 10)$ all become $(3, 0)$. You have lost all information

about the original y-component, which is why the operation cannot be reversed.

Singular Matrices & Degenerate Transformations

When transformations lose information



Original vectors (red) are all projected onto the x-axis, losing their y-component. All become $(x, 0)$ (blue).

Shearing: The Skewing Effect

- A **shearing transformation** (or skewing) is a linear transformation that distorts a shape by sliding all points parallel to a fixed axis.
- Key Property: It **preserves the area** of the object being transformed.

The Transformation Matrix

A point $\mathbf{P}(x, y)$ is transformed to $\mathbf{P}'(x', y')$ using a shear factor S.

1. X-Shear (Parallel to X-axis)

- y' remains y.
- x' depends on y: $x' = x + S_x \cdot y$

$$\mathbf{M_x} = \begin{bmatrix} 1 & S_x \\ 0 & 1 \end{bmatrix}$$

2. Y-Shear (Parallel to Y-axis)

- x' remains x.
- y' depends on x: $y' = S_y \cdot x + y$

$$\mathbf{M_y} = \begin{bmatrix} 1 & 0 \\ S_y & 1 \end{bmatrix}$$

Numerical Example: X-Shear on a Square

Task: Apply an X-Shear with factor $S_x = 2$ to a unit square.

- **Original Vertices (Unit Square):** $P_1(0,0)$, $P_2(1,0)$, $P_3(1,1)$, $P_4(0,1)$
- **X-Shear Formula:** $x' = x + 2y, \quad y' = y$

$$\mathbf{M_x} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Table: Transformation Calculation ($S_x = 2$)

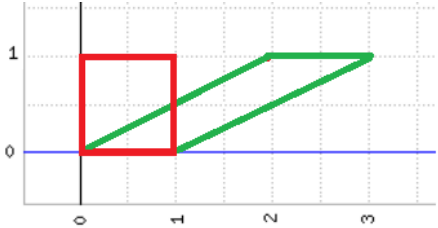
Point	**Original** (x, y)	**Transformed** (x', y')
P_1	$(0,0)$	$(0 + 2 \cdot 0, 0) = (0,0)$
P_2	$(1,0)$	$(1 + 2 \cdot 0, 0) = (1,0)$
P_3	$(1,1)$	$(1 + 2 \cdot 1, 1) = (3,1)$
P_4	$(0,1)$	$(0 + 2 \cdot 1, 1) = (2,1)$

Numerical Example: X-Shear on a Square

Table: Transformation Calculation ($S_x = 2$)

Point	**Original** (x, y)	**Transformed** (x', y')
P_1	$(0, 0)$	$(0 + 2 \cdot 0, 0) = (0, 0)$
P_2	$(1, 0)$	$(1 + 2 \cdot 0, 0) = (1, 0)$
P_3	$(1, 1)$	$(1 + 2 \cdot 1, 1) = (3, 1)$
P_4	$(0, 1)$	$(0 + 2 \cdot 1, 1) = (2, 1)$

Visual Result:



The X-Shear shifts the top edge (P3, P4) sideways.

Matrices & Operations

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12. Matrices as Data Structures

- A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ has m rows and n columns.
- **In AI:**
  - Rows often represent **data samples** (observations).
  - Columns often represent **features** (variables).
- Example: A dataset of m patients with n vital signs.

13. Matrix Addition and Scalar Multiplication

- **Addition:** Requires matrices to have the same dimensions. Performed element-wise.
$$\mathbf{A} + \mathbf{B} = \mathbf{C} \quad \text{where } c_{ij} = a_{ij} + b_{ij}$$
- **Scalar Multiplication:** Multiplies every element in the matrix by a scalar c.

14. Matrix Transpose

- The transpose of $\mathbf{A}$ ($m \times n$) is denoted $\mathbf{A}^T$ ($n \times m$).
- Rows become columns, and columns become rows: $a_{ij}^T = a_{ji}$.
- **Symmetric Matrix:** A square matrix where $\mathbf{A} = \mathbf{A}^T$.
- **AI Context:** Essential for defining covariance matrices.

15. Matrix Multiplication (The Core Operation)

- For $\mathbf{A} \in \mathbb{R}^{m \times k}$ and $\mathbf{B} \in \mathbb{R}^{k \times n}$, the result $\mathbf{C} = \mathbf{AB}$ is $\mathbb{R}^{m \times n}$.
- **Rule:** The number of columns in **A** must equal the number of rows in **B**.
- c_{ij} is the dot product of **A**'s i-th row and **B**'s j-th column.

16. Matrix Multiplication in Neural Networks

- The central operation in a feed-forward layer:

$$\mathbf{a}^{(l)} = \sigma(\mathbf{W}^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)})$$

- $\mathbf{W}^{(l)}$ is the weight matrix.
- $\mathbf{a}^{(l-1)}$ is the input vector (activation from the previous layer).
- This multiplication represents applying a linear transformation to the input data.

17. Properties of Matrix Multiplication

- **Associative:** $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.
- **Distributive:** $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.
- **NOT Commutative:** $\mathbf{AB} \neq \mathbf{BA}$ in general.

18. Special Matrices (Identity)

- **Identity Matrix (I):** A square matrix with ones on the main diagonal and zeros elsewhere.
- It acts like the number 1 in multiplication: $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$.
- The Identity Matrix represents the "do nothing" transformation.

19. Special Matrices (Diagonal and Symmetric)

- **Diagonal Matrix (D):** Non-zero entries only on the main diagonal ($d_{ij} = 0$ if $i \neq j$).
- **Symmetric Matrix (A):** $\mathbf{A} = \mathbf{A}^T$.
- **AI Context:** Covariance matrices are always symmetric. Diagonal matrices simplify calculations involving eigenvalues.

20. Matrices as Linear Transformations

- Multiplying a vector $\mathbf{x}$ by a matrix $\mathbf{A}$ transforms $\mathbf{x}$ to a new vector $\mathbf{y} = \mathbf{A}\mathbf{x}$.
- This transformation can involve rotation, scaling, shearing, or projection.
- Understanding these transformations is key to interpreting what a neural network layer is "learning."

Linear Systems & Elimination

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22. Systems of Linear Equations

- A set of m linear equations in n variables:

$$a_{11}x_1 + \cdots + a_{1n}x_n = b_1$$

$$\cdots$$

$$a_{m1}x_1 + \cdots + a_{mn}x_n = b_m$$

- In matrix form: $\mathbf{Ax} = \mathbf{b}$.
- Finding the solution $\mathbf{x}$ is a core computational task.

What is the Determinant?

- **Definition:** The determinant is a **scalar value** calculated from the elements of a **square matrix** ($n \times n$).
- **Notation:** It is denoted as $\det(A)$ or by vertical bars $|A|$.
- **Key Role:** It is the primary indicator of a matrix's fundamental properties, particularly its **invertibility** and its **geometric scaling effect**.

Central Question

Is the matrix A invertible?

$$\text{Yes, if } \det(A) \neq 0. \qquad \text{No, if} \det(A) = 0.$$

Calculation: The $\mathbf{2} \times \mathbf{2}$ Case

Simple Cross-Multiplication

1. The formula is straightforward: Multiply the elements on the main diagonal and subtract the product of the elements on the anti-diagonal.

Formula

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\det(A) = ad - bc$$

Numerical Example

$$A = \begin{pmatrix} 5 & 2 \\ 4 & 3 \end{pmatrix}$$

$$\det(A) = (5)(3) - (2)(4)$$

$$\det(A) = 15 - 8$$

$$\det(A) = 7$$

Conclusion

Since $\det(A) = 7 \neq 0$, the matrix A is **invertible**.

Calculation: The $\mathbf{3} \times \mathbf{3}$ Case

Using Cofactor Expansion (or Sarrus' Rule)

1. Calculation involves breaking down the 3×3 matrix into 2×2 sub-determinants (minors).
2. We use Cofactor Expansion along the first row: $a_{11} \cdot C_{11} - a_{12} \cdot C_{12} + a_{13} \cdot C_{13}$

Matrix B

$$B = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}$$

Step-by-Step

$$\begin{aligned} \det(B) &= 1 \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} - 2 \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} \\ &= 1(1 \cdot 1 - 2 \cdot 1) - 2(3 \cdot 1 - 2 \cdot 0) + 0(\dots) \\ &= 1(1-2) - 2(3-0) + 0 \\ &= -1 - 6 \\ &= -7 \end{aligned}$$

Conclusion

Since $\det(B) = -7 \neq 0$, the matrix B is also **invertible**.

Significance: Geometry and Independence

The True Meaning of $\det(A)$

1. **Linear Independence:**
   - If $\det(A) = 0$, the transformation collapses the space onto a lower dimension (e.g., squishes a 3D cube onto a 2D plane).
   - This means the column vectors of A are **linearly dependent**.

Example of $\det(C) = 0$

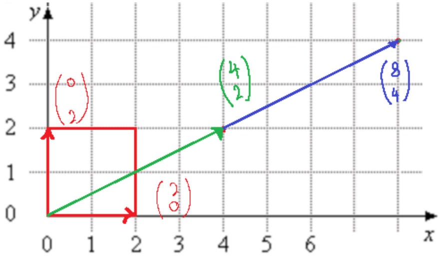
$$C = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$$

$$\det(C) = (2)(2) - (4)(1) = 4 - 4 = 0$$

Numerical Example

$$Z = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix} \times \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 8 \\ 2 & 4 \end{pmatrix}$$

Result: C is **not invertible**. The second column is exactly twice the first ($4 = 2 \times 2$ and $2 = 2 \times 1$), so the columns are dependent.



Geometric Significance: Area Scaling Factor

How the Determinant Changes Area in 2D

1. **Geometric Scaling:**
   - The absolute value $|\det(A)|$ represents the **scaling factor** of the area (in 2D) or volume (in 3D) after the linear transformation A is applied.
   - The **sign** indicates orientation: Positive means orientation is preserved; Negative means orientation is reversed (a reflection).

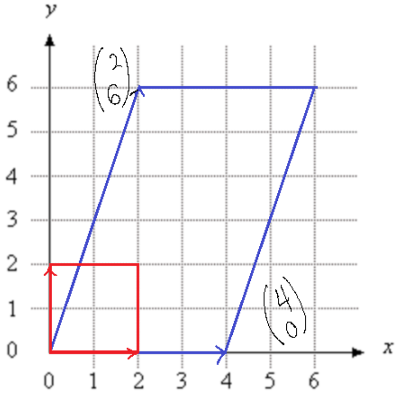
Example of $\det(A) > 0$

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

$\det(A) = (2)(3) - (1)(0) = 6 - 0 = \mathbf{6}$

Numerical Example

$$Z = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \times \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 0 & 6 \end{pmatrix}$$



Geometric Significance: Area Scaling Factor

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   - The **sign** indicates orientation: Positive means orientation is preserved; Negative means orientation is reversed (a reflection).

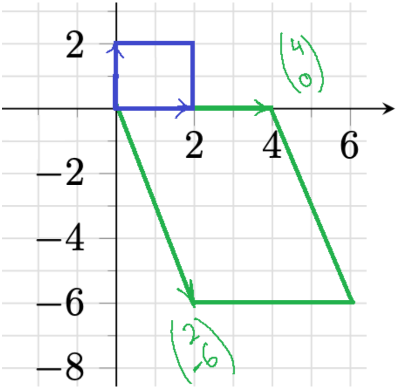
Example of $\det(A) < 0$

$$A = \begin{pmatrix} 2 & 1 \\ 0 & -3 \end{pmatrix}$$

$$\det(A) = (-2)(3) - (1)(0) = \mathbf{-6}$$

Numerical Example

$$Z = \begin{pmatrix} 2 & 1 \\ 0 & -3 \end{pmatrix} \times \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 0 & -6 \end{pmatrix}$$



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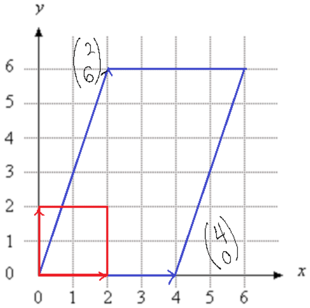
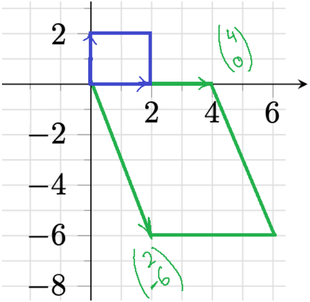
How the Determinant Changes Area in 2D





Geometric Conclusion

The area of the resulting parallelogram is exactly equal to the absolute value of the determinant:

$$\text{New Area} = |\det(A)| \times \text{Initial Area} = 6 \times 4 = \mathbf{24}$$

The determinant is the **Area Magnification Factor**.